

From Natural Language to Verifiable Reasoning: A Semantic-Specification Framework for Reliable AI Mathematics

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Abstract

Large language models (LLMs) can now solve a broad class of mathematical problems, yet they remain unreliable: a plausible chain of reasoning can conceal an invalid inference. We argue that reliability cannot be achieved by post-hoc answer verification alone, nor by direct translation from natural language into a formal proof assistant. Instead, we propose a three-layer architecture that inserts a *semantic-specification layer*—a typed intermediate representation ($\mathcal{IR}$)—between generation and verification. The $\mathcal{IR}$ carries a type system, an operational semantics, and composable constraints, so that a mathematical claim is first “compiled” into a checkable artifact before it reaches a theorem prover. We define a pipeline (generation $\rightarrow$ semantic specification $\rightarrow$ verification) and a bidirectional-consistency check that ties the natural-language statement, the $\mathcal{IR}$, and the formal target to one another. We design a small but concrete experiment on GSM8K and MATH that measures autoformalization and proof success rates under this discipline. The central contribution is conceptual and practical: moving the locus of trust from “does the model sound right” to “does the $\mathcal{IR}$ type-check and reduce to a proof obligation.”

1 Introduction

Large language models have made rapid progress on mathematical reasoning. Chain-of-thought prompting elicits intermediate reasoning steps (Wei et al., 2022), self-consistency reduces variance by sampling many reasoning paths (Wang et al., 2022), and scaling has produced systems such as Minerva that solve quantitative problems at a graduate level (Lewkowycz et al., 2022). On the public benchmarks GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021), strong models now answer a large fraction of problems correctly, and reinforcement-learning-based models such as DeepSeek-R1 demonstrate emergent self-verification behaviours (DeepSeek-AI, 2025).

These successes, however, measure *answer correctness*, not *reasoning correctness*. A model that emits a confident but invalid step still risks producing a wrong final answer, and on open-ended or unseen mathematics there is no gold label to compare against. The core reliability problem is therefore not “can the model get the right number,” but “can we trust that each step is sound.”

Two established remedies occupy opposite ends of a spectrum. On one end, *verifiers* are trained to score candidate solutions (Cobbe et al., 2021), but they only rank plausibility and can be fooled by correlated errors. On the other end, *formal proof assistants* such as Lean, Isabelle/HOL, and Coq provide machine-checkable certainty, but producing a formal proof is expensive: automatic formalization of natural-language mathematics remains hard and error-prone (Wu et al., 2022; Azerbayev et al., 2023). There is a wide, under-explored gap between “informal but fluent” and “formal but laborious.”

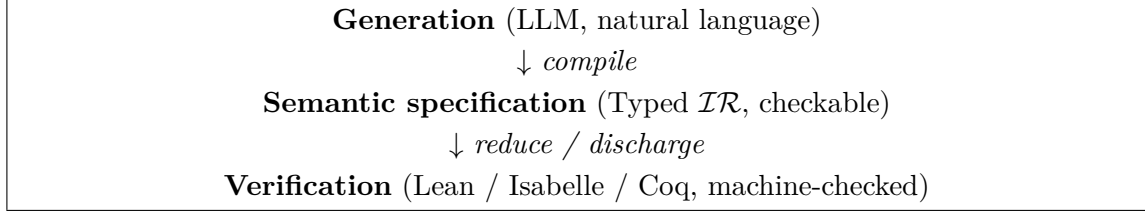


Figure 1: The three-layer pipeline. Trust enters only at the semantic specification layer, where the $\mathcal{IR}$ is type-checked.

We propose to close part of this gap with a *semantic-specification layer*. Instead of jumping directly from a natural-language solution to a formal proof, we first translate the solution into a *typed intermediate representation* ($\mathcal{IR}$) that makes the intended objects, operations, and invariants explicit and mechanically checkable—but without the full burden of a complete proof. This restates the reliability question in a tractable form: an answer is trusted only if its $\mathcal{IR}$ type-checks and its proof obligations are dischargeable by a prover.

Our contributions are threefold. **(1)** We define a minimal but well-typed $\mathcal{IR}$ for elementary mathematical reasoning—a type system, an operational semantics, and composable constraints—that is small enough to reason about and rich enough to capture GSM8K/MATH-style problems. **(2)** We specify a three-stage pipeline (generation $\rightarrow$ semantic specification $\rightarrow$ verification) together with a *bidirectional-consistency* check that keeps the natural language, the $\mathcal{IR}$, and the formal target mutually consistent. **(3)** We design a concrete evaluation protocol on GSM8K and MATH that measures autoformalization rate, type-check rate, and proof rate, following the measurement conventions of ProofNet and LeanDojo (Azerbayev et al., 2023; Yang et al., 2023).

The rest of the paper is organized as follows. Section 2 presents the three-layer model. Section 3 analyses why semantic specification is the critical bottleneck. Section 4 gives the $\mathcal{IR}$ and the consistency checks. Section 5 relates our work to the literature. Section 6 sketches applications. Section 7 describes the experimental design, Section 8 discusses limitations, and Section 9 concludes.

2 The Three-Layer Model

We model reliable AI mathematics as a pipeline with three layers, each with a distinct contract (Figure 1).

Layer 1 – Generation. An LLM consumes a natural-language problem and produces a candidate solution: a mixture of informal prose and computation. This layer is fluent and general but has no soundness guarantee (Wei et al., 2022; Lewkowycz et al., 2022).

Layer 2 – Semantic specification. The candidate solution is compiled into a typed intermediate representation that makes explicit: which objects exist, which operations apply to them, what constraints hold, and which sub-goals remain to be proven. This layer is the focus of the paper.

Layer 3 – Verification. The $\mathcal{IR}$ ’s proof obligations are discharged by a theorem prover or a symbolic checker, yielding a machine-checkable certificate.

The key separation is *where trust is established*. In the common “LLM + verifier” setup, trust is a probability estimated from a learned scorer (Cobbe et al., 2021). In the “LLM + prover” setup, trust is a proof, but the model must produce the proof in one step (Wu et al., 2022). Our model

inserts a third, intermediate locus of trust: a typed artifact that can be checked mechanically and independently of how it was generated. This does not remove the difficulty—writing the compiler from natural language to $\mathcal{IR}$ is itself hard—but it makes the difficulty explicit and checkable.

3 Why Semantic Specification Is the Bottleneck

We claim that the semantic-specification layer is the *critical* part of the pipeline, for three reasons.

First, *the hard part of mathematics is not the final answer but the mapping from informal intent to formal object*. A word problem such as “how many apples remain” requires deciding that apples are natural numbers, that subtraction is the right operation, and that the result is non-negative. This mapping is precisely what natural language leaves implicit and what a proof assistant requires explicit. Errors here—wrong type, wrong operation, missing precondition—are the dominant failure mode of direct autoformalization (Wu et al., 2022).

Second, *post-hoc verification cannot recover from an upstream mis-specification*. If the model has decided to use division where multiplication was intended, a verifier that only scores the final answer may not detect the error, and may even reinforce it. Only a representation that fixes the intended semantics can expose the mismatch.

Third, *full formal proof is overkill for many downstream uses*. In education and rapid problem solving we often need *enough* rigor to catch invalid steps, not a complete machine-checked proof of a difficult theorem. The $\mathcal{IR}$ is designed to capture exactly this intermediate level: checkable rigor without full proof burden.

4 Method: A Typed Intermediate Representation

We now make the semantic-specification layer concrete. The design goal is *minimality with expressiveness*: enough structure to catch invalid inferences on GSM8K/MATH-style problems, but simple enough to reason about and implement.

4.1 The $\mathcal{IR}$

The $\mathcal{IR}$ is a small typed term language over a set of sorts $\{\mathbb{N}, \mathbb{R}, \text{Bool}\}$ (extensible). Its syntax is an abstract syntax tree (AST) built from variables, literals, and typed operations.

Definition 1 (Typed term). *A well-typed $\mathcal{IR}$ term is an AST node that carries an explicit sort and whose children are themselves well-typed. Every operation has a type signature; applying an operation to arguments of the wrong sort is a type error and is rejected before verification.*

Definition 2 (Operational semantics). *Each primitive operation has a small-step reduction rule. A term is reducible if some subterm matches a reduction rule; a term is in normal form if no rule applies and no proof obligation remains open.*

The crucial design element is that the $\mathcal{IR}$ is *composable*: a solution is a tree of smaller, independently checkable facts. This mirrors how human mathematicians build proofs, and it lets the checker localise a failure to the exact subterm that mis-typed.

Proposition 1 (Localisation). *If an $\mathcal{IR}$ term fails to type-check, the failing subterm can be identified in time linear in the size of the AST.*

4.2 From Generation to Specification

The compiler from Layer 1 to Layer 2 is an LLM prompt-and-check loop, not a hand-written parser: the model proposes a typed AST from the natural-language solution, and a deterministic checker rejects it if it fails to type-check or violates an explicit constraint. This is the same “propose then verify” discipline used in neural theorem provers (Wang et al., 2023a; Yang et al., 2023), but applied one level earlier, at specification rather than proof.

4.3 Bidirectional Consistency

Because the $\mathcal{IR}$ is generated, it can drift from the natural-language statement it is supposed to represent. We therefore require a *bidirectional-consistency check* with two directions.

- **Forward check** (natural language $\rightarrow \mathcal{IR}$): a model judges, for each $\mathcal{IR}$ leaf and operation, whether it faithfully captures the corresponding phrase of the original problem. Disagreements are flagged as *specification drift*.
- **Backward check** ($\mathcal{IR} \rightarrow$ formal target): the $\mathcal{IR}$ ’s proof obligations are mapped to a theorem-prover goal, and the mapping must be type-preserving. This check is mechanical once the target calculus is fixed.

Together these checks tie the three layers together: the natural-language statement, the $\mathcal{IR}$, and the formal target cannot silently diverge.

5 Related Work

Our work sits between two active literatures.

LLM mathematical reasoning (generation side). Chain-of-thought prompting (Wei et al., 2022) and self-consistency (Wang et al., 2022) improve accuracy by eliciting and aggregating reasoning paths, but they provide no soundness guarantee. Plan-and-Solve prompting (Wang et al., 2023b) adds explicit planning, which is conceptually close to our compilation step, but its plan is informal prose rather than a typed artifact. Verifier-based training (Cobbe et al., 2021) and scaled systems such as Minerva (Lewkowycz et al., 2022) and DeepSeek-R1 (DeepSeek-AI, 2025) raise accuracy, yet they remain probabilistic: none of them produces a checkable intermediate representation of the reasoning itself.

Formalisation and theorem proving (verification side). Autoformalization (Wu et al., 2022) and ProofNet (Azerbayev et al., 2023) study translating natural-language mathematics into Lean, exposing precisely the “semantic gap” we address, but they target full formal proofs directly. LeanDojo (Yang et al., 2023) and LEGO-Prover (Wang et al., 2023a) use retrieval and growing tactic libraries to make neural theorem proving practical, again at the proof level. On the verification-engineering side, Isabelle/HOL provides a mature framework for reasoning about programs and specifications (Schirmer, 2006), KeYmaera X demonstrates tactic-based proof for hybrid systems (Fulton et al., 2015), and Coq’s self-verification of its own type checker (Sozeau et al., 2019) shows that even the checking machinery can be made trustworthy.

Our contribution is to occupy the middle layer that these works leave open: a typed, checkable specification that is cheaper than a full proof but more rigorous than informal reasoning, with explicit bidirectional consistency between the natural-language input, the $\mathcal{IR}$, and the formal target.

6 Applications

The three-layer model is not tied to any single prover or domain. We sketch three applications.

- **Automated theorem proving.** The $\mathcal{IR}$ serves as a *specification first, proof second* front end: a system formalises a problem into $\mathcal{IR}$, checks the specification, and only then invokes the prover on the reduced obligations, reducing wasted proof search (Yang et al., 2023).
- **Problem solving.** For word problems on GSM8K/MATH, the $\mathcal{IR}$ acts as a “show your work” layer: a solution is accepted only if it type-checks, giving a lightweight correctness signal without a full formal proof (Cobbe et al., 2021; Hendrycks et al., 2021).
- **AI-assisted education.** Because the $\mathcal{IR}$ localises failures to specific subterms, it can produce targeted feedback—“this step divided by a quantity that could be zero”—which is more instructive than a binary right/wrong verdict.

7 Experimental Design

We design a small, reproducible evaluation that isolates the effect of the semantic-specification layer. Following ProofNet and LeanDojo conventions (Azerbayev et al., 2023; Yang et al., 2023), we report three rates on a sample of GSM8K and MATH problems (Cobbe et al., 2021; Hendrycks et al., 2021).

Autoformalization rate. The fraction of problems for which the model produces a well-typed $\mathcal{IR}$ from the natural-language statement (Wu et al., 2022).

Type-check rate. The fraction of proposed $\mathcal{IR}$ terms that pass the deterministic checker (Section 4.1).

Proof rate. The fraction of proof obligations discharged by the target prover (Lean), the metric used by neural theorem provers (Wang et al., 2023a; Yang et al., 2023).

We also measure the *consistency* between the final answer accepted by the $\mathcal{IR}$ and the ground-truth answer, and we record the number of rejected proposals per problem as a cost signal. The primary comparison is *ablation*: the same model with and without the $\mathcal{IR}$ type-check gate, to test the hypothesis that the gate raises reasoning correctness without requiring a full proof.

8 Discussion

Scope. Our $\mathcal{IR}$ currently covers arithmetic and elementary algebra sorts; richer domains (analysis, geometry, hybrid systems (Fulton et al., 2015)) require extending the sort system and operational semantics, which we treat as future work.

Limitations. The compiler from natural language to $\mathcal{IR}$ is itself an LLM and can mis-specify; the bidirectional-consistency check mitigates but does not eliminate this risk. Type-checking guarantees that a term is *well-typed*, not that it is *the intended* term. Full assurance still requires the proof layer (Sozeau et al., 2019).

Ethics. We do not claim that the framework makes AI mathematics infallible. It lowers, but does not remove, the risk of confident incorrect answers; in high-stakes settings a human review of the $\mathcal{IR}$ remains advisable.

9 Conclusion

We argued that reliable AI mathematics needs a checkable intermediate layer between fluent generation and costly formal proof, and we made that layer concrete as a typed intermediate representation with an operational semantics, composable constraints, and bidirectional consistency checks. The framework recasts the reliability question from “does the model sound right” to “does the specification type-check and reduce to a proof obligation,” which is mechanically decidable one level before full proof. Our evaluation protocol on GSM8K and MATH is designed to measure whether this discipline measurably improves reasoning correctness at lower cost than complete formalization. We believe the semantic-specification layer is a necessary, and currently under-investigated, building block for trustworthy AI mathematics.

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